



FINAL YEAR ENGINEERING PROJECT

Smart Medicine Availability & Intelligent Janaushadhi Recommendation System

FRONTEND BACKEND MAPPING

Project	Smart Medicine Availability & Intelligent Janaushadhi Recommendation System
Document	FRONTEND BACKEND MAPPING
Version	v1.0.0
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Complete API Integration Reference

Every frontend action mapped to its backend endpoint.

Authentication

Frontend Page	Component/Action	HTTP Method	Endpoint	Request Body	Expected Response
LoginPage	form submit	POST	/api/v1/auth/login	{ email, password }	{ user: {...}, accessToken, refreshToken }
RegisterPage	form submit	POST	/api/v1/auth/register	{ fullName, email, mobile, password }	{ message: "OTP sent" }
VerifyOtpPage	OTP submit	POST	/api/v1/auth/verify-otp	{ email, otp }	{ message: "Verified" }
VerifyOtpPage	Resend OTP	POST	/api/v1/auth/resend-otp	{ email }	{ message: "OTP resent" }
ForgotPasswordPage	form submit	POST	/api/v1/auth/forgot-password	{ email }	{ message: "Reset OTP sent" }
ResetPasswordPage	form submit	POST	/api/v1/auth/reset-password	{ email, otp, newPassword }	{ message: "Password reset" }
Any page	token refresh	POST	/api/v1/auth/refresh-token	{ refreshToken }	{ accessToken, refreshToken }
TopBar	logout button	POST	/api/v1/auth/logout	(empty)	{ message: "Logged out" }

User Profile

Frontend Page	Component/Action	HTTP Method	Endpoint	Request	Expected Response
ProfilePage	page load	GET	/api/v1/users/me	(headers only)	{ id, name, email, mobile, role, avatar, ... }
ProfilePage	save changes	PUT	/api/v1/users/me	{ name, email, mobile }	{ updated user object }
ProfilePage	delete account	DELETE	/api/v1/users/me	(empty)	{ message: "Account deleted" }
TopBar	user info	GET	/api/v1/users/me	(headers only)	{ name, avatar, role }

Medicine Search

Frontend Page	Component/Action	HTTP Method	Endpoint	Request Params	Expected Response
MedicineSearchPage	search input	GET	/api/v1/medicines/search	{ query }	{ [{ id, name, genericName, ... }] (suggestions) }
MedicineSearchPage	page load	GET	/api/v1/medicines/popular		{ [{ id, name, ... }] }
MedicineSearchPage	page load	GET	/api/v1/medicines/categories		{ [{ id, name, icon }] }
SearchResultsPage	search results	GET	/api/v1/medicines/search	{ query }&type=generic&sort=price&page=2&limit=10	{ medicines: [...], totalPages: 2 }
MedicineDetailsPage	page load	GET	/api/v1/medicines/:id		{ id, name, genericName, composition, mobile, ... }
MedicineDetailsPage	alternatives section	GET	/api/v1/medicines/alternatives/:id		{ [{ id, name, price, isJanAushadhi, ... }] }
Admin — AdminMedicines	list medicines	GET	/api/v1/medicines	?page=1&limit=10	{ medicines: [...], total }
Admin — AdminMedicines	create medicine	POST	/api/v1/medicines	{ name, genericName, composition }	{ created medicine }
Admin — AdminMedicines	update medicine	PUT	/api/v1/medicines/:id	{ updated fields }	{ updated medicine }

Frontend Page	Component/Action	HTTP Method	Endpoint	Request Params	Expected Response
Admin — AdminMedicines	delete medicine	DELETE	/api/v1/medicines/:id—		{ message: "Deleted" }

Generic Recommendation

Frontend Page	Component/Action	HTTP Method	Endpoint	Request	Expected Response
GenericRecommendationPage	load	GET	/api/v1/medicines/:id/generic-recommendations		{ medicine: { ... }, generics: [{ ... }], savings: { ... } }
GenericRecommendationPage	save recommendation	POST	/api/v1/users/me/saved-medicines/:id		{ message: "Saved" }

Pharmacy / Nearby

Frontend Page	Component/Action	HTTP Method	Endpoint	Request Params	Expected Response
NearbyPharmaciesPage	load	GET	/api/v1/pharmacies/nearby	{ lat, lng, radius }	{ id, name, address, lat, lng, distance, isMedicineInStock }
NearbyPharmaciesPage	pharmacy search	GET	/api/v1/pharmacies/search	{ name }	{ id, name, address, ... }
MedicineDetailsPage	nearby section	GET	/api/v1/pharmacies/nearby	{ lat, lng, radius }	{ id, name, address, lat, lng, distance, isMedicineInStock }
Admin — AdminPharmacies	load	GET	/api/v1/pharmacies	{ page=1&limit=10 }	{ pharmacies: [...], total }
Admin — AdminPharmacies	add pharmacy	POST	/api/v1/pharmacies	{ name, address, lat, lng, telephone }	{ id }
Admin — AdminPharmacies	edit pharmacy	PUT	/api/v1/pharmacies/:id	{ updated fields }	{ updated pharmacy }

Pharmacy Inventory

Frontend Page	Component/Action	HTTP Method	Endpoint	Request	Expected Response
InventoryPage	table load	GET	/api/v1/inventory	{ search={q}&page={p}&limit={l} }	{ items: [...], total }
MedicineFormPage	add submit	POST	/api/v1/inventory	{ medicineId, quantity, expiryDate, batchNo }	{ id }
MedicineFormPage	edit submit	PUT	/api/v1/inventory/:id	{ updated fields }	{ updated stock item }
InventoryPage	delete row	DELETE	/api/v1/inventory/:id—		{ message: "Deleted" }
MedicineFormPage	load for edit	GET	/api/v1/inventory/:id—		{ id, medicine, quantity, price, expiryDate }

User Dashboard Data

Frontend Component	HTTP Method	Endpoint	Expected Response
Dashboard summary	GET	/api/v1/users/me/dashboard	{ recentSearches, savedMedicines, stats }
Recent notifications	GET	/api/v1/users/me/notifications	{ limit, type, title, message, time, isRead }
Saved medicines	GET	/api/v1/users/me/saved-medicines	{ id, medicine, savedAt }

Notifications

Frontend Page	Action	HTTP Method	Endpoint	Request	Expected Response
NotificationsPage	load all	GET	/api/v1/users/me/notifications	{?page=1&limit=10&role=patient}	{ id, type, title, message, time, isRead }
NotificationsPage	mark one read	PATCH	/api/v1/users/me/notifications/:id/read	{ message: "Updated" }	
NotificationsPage	mark all read	PATCH	/api/v1/users/me/notifications/read-all	{ message: "All marked read" }	
NotificationsPage	dismiss	DELETE	/api/v1/users/me/notifications/:id		{ message: "Dismissed" }

Admin User Management

Frontend Page	Action	HTTP Method	Endpoint	Request	Expected Response
AdminUsers	load table	GET	/api/v1/users	{?page=1&limit=10&role=patient}	{ id, name, email, role, createdAt, ... }
AdminUsers	view user	GET	/api/v1/users/:id	—	{ id, name, email, role, createdAt, ... }
AdminUsers	update role	PUT	/api/v1/users/:id	{ role }	{ updated user }
AdminUsers	delete user	DELETE	/api/v1/users/:id	—	{ message: "Deleted" }

Admin Generic Mapping

Frontend Page	Action	HTTP Method	Endpoint	Request	Expected Response
AdminGenericMapping	load mappings	GET	/api/v1/generic-mappings	{?page=1}	{ { brandedMedicineId, genericMedicineId, ... } }
AdminGenericMapping	add mapping	POST	/api/v1/generic-mappings	{ brandedId, genericId }	{ created mapping }
AdminGenericMapping	delete mapping	DELETE	/api/v1/generic-mappings/:id		{ message: "Deleted" }

Analytics

Frontend Page	Action	HTTP Method	Endpoint	Expected Response
AdminAnalytics	load data	GET	/api/v1/analytics/supplies	{ totalSearches, totalUsers, avgSavings, ... }
AdminDashboard	stats	GET	/api/v1/admin/dashboard	{ users, pharmacies, medicines, searches }
AdminReports	generate	POST	/api/v1/reports/generate	{ type, dateRange, downloadUrl }

Request / Response Headers

Every authenticated request must include:

```
Authorization: Bearer {accessToken}
Content-Type: application/json
```

This is handled automatically by `axiosClient.js` interceptors.

Standard success response shape:

```
{
  "success": true,
  "data": { ... },
  "message": "Success"
}
```

Standard error response shape:

```
{
  "success": false,
  "message": "Error description",
  "error": "SPECIFIC_ERROR_CODE"
}
```

Pagination response shape:

```
{
  "data": [...],
  "total": 150,
  "page": 1,
  "totalPages": 15,
  "limit": 10
}
```